Q1. In wide-band FM transmission, a pre-emphasis process is applied in the transmitter. The pre-emphasis process applies a gain to the signal components which are higher than a certain frequency (say f_{pc} Hz). In the receiver, this process is reversed (de-emphasis) by applying an attenuation to the signal components which are higher than the frequency f_{pc} Hz. The pre-emphasis and de-emphasis are implemented using filters. Fig. 4 (a) shows the pre-emphasis filter response. Due to the wrong configuration in the receiver, the receiver applies the de-emphasis filter response given in Fig. 4 (b). Here, assume that the transmit signal is in the frequency range of 0 Hz to $2f_{pc}$ Hz and the amplitudes of all frequency components of the transmitted signal $s(t)$ (before the pre-emphasis filter) is 0 dB. What would be the output spectrum after the de-emphasis filter ($y(t)$) (see Fig. 5)?. Here, ignore the impact of the wireless channel.
Hint:- $log(a/b) = log\ (a)\ -\ log\ (b)\ and\ log(a\ x\ b) = log\ (a)\ + log\ (b)$.

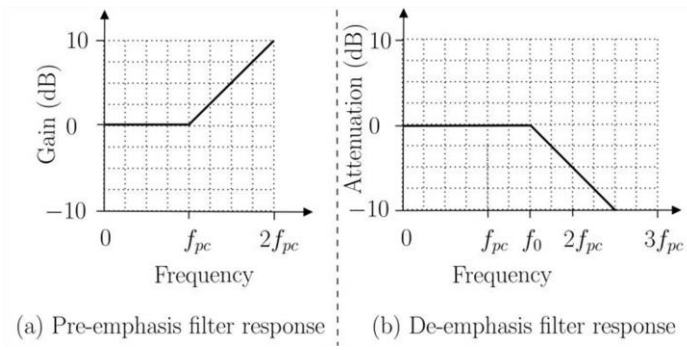


Fig. 4 Pre-emphasis and de-emphasis filter responses.

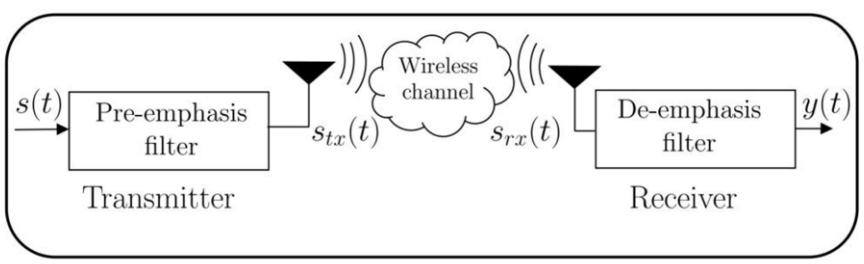


Fig. 5 Pre-emphasis and de-emphasis process.

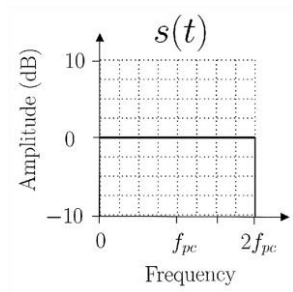
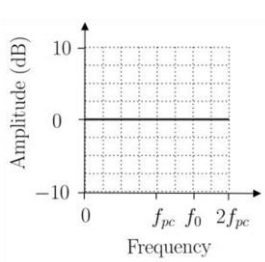
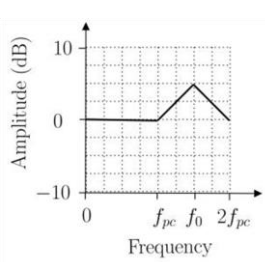


Fig. 6 $s(t)$

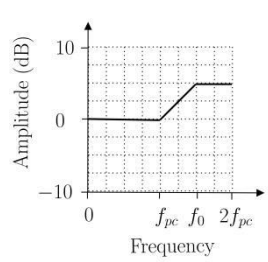
(1)



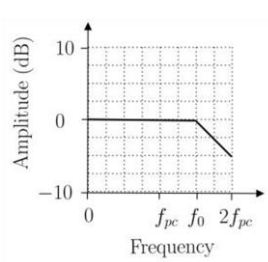
(2)



(3)



(4)



Q2. Timing synchronization is one of the fundamental requirements for the proper functioning of any wireless communication system. Early/Late Gate Time Synchronization is a basic approach that can be used for timing synchronization. Fig. 8(a) shows a block diagram of early/late gate time synchronization. A reference wave which is generated by the waveform generator is used to determine the timing error. The value given by Q is used to calculate the time difference between the transmitter and receiver. For the received signal given in Fig. 8 (b), what is the value of Q ? (Hint: - output value after the Magnitude block is equal to the area of the input signal to the integrator block until time T.)

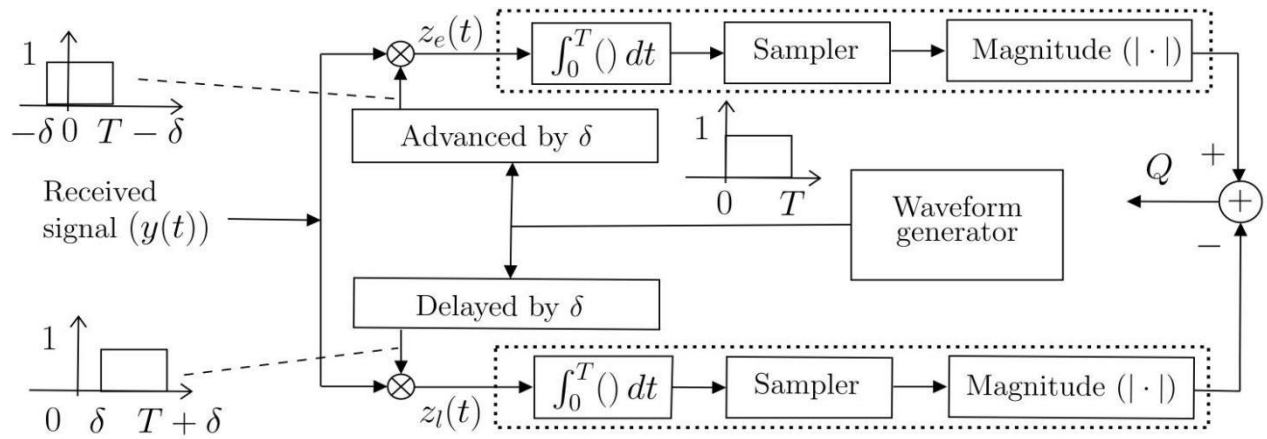


Fig. 8 (a) Early/Late Gate Time Synchronization

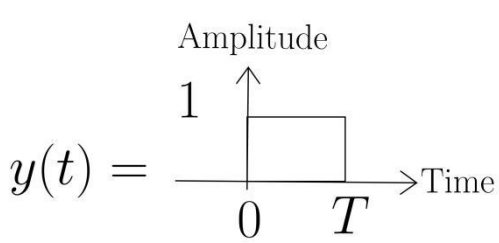


Fig. 8 (b) Received signal ($y(t)$)

(1) δ (2) 2δ (3) 0 (4) T-δ

Q3. Consider the Early/Late Gate Time Synchronization block given in question 2. For the received signal given in Fig. 9, what is the value of Q ?. Here $\epsilon < \delta$.

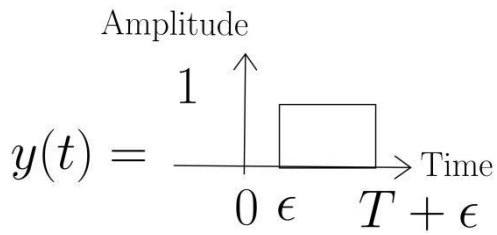


Fig. 9 Received signal ($y(t)$)

(1) $\delta - \epsilon$ (2) 2δ (3) 2ϵ (4) -2ϵ

Q4. Consider the Early/Late Gate Time Synchronization block given in question 2. For the received signal given in Fig. 10, what is the value of Q ? Here $\epsilon < \delta$.

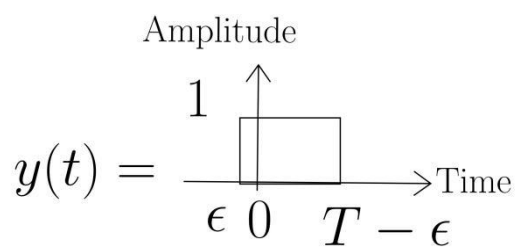


Fig. 10 Received signal ($y(t)$)

(1) $\delta - \epsilon$ (2) 2δ (3) 2ϵ (4) -2ϵ